

Pipelining

- The MIPS Pipeline
 - 5 stages - instruction, execution, memory, align, and writeback
 - Instruction fetch
 - Instruction is fetched from memory
 - Execution
 - Only time memory is accessed is with a load or a store
 - Multiply/divide operations are started, they are the longest operations to do
 - Memory
 - Load and store are performed
 - Align
 - Multiply/divide complete
 - Align data to word boundary
 - Writeback
 - Register loaded w register-to-register or load value
- Reducing Effects of Branching
 - MIPS and ARM use pipelining heavily
 - Multiple Streams
 - Have two pipelines and go down both paths
 - Once outcome is determine, go down the correct pipeline
 - Requires twice as many resources
 - Some resources are not shareable
 - How many duplicate resources are needed if there are nested branches?
 - Pre-fetch Branch Target
 - Target is fetched file pipe executives instructions following branch
 - Keep target until branch is executed
 - Used by IBM 360/91
 - Loop Buffer
 - Specialize cache
 - FIFO arrangement
 - Each line in the buffer contains addressing information and possibly decoding information
 - Gives a history of execution, if a loop is encountered, the code to execute is already there
 - Loop buffers are pretty small, if a loop is large, then it doesn't help
 - Branch Prediction
 - Static branch prediction
 - Doesn't really "predict" - assumes that branches are never taken, always taken, or based on the op code
 - Predict by opcode is helpful - a loop is typically taken
 - Predict never taken
 - Assume that the jump will not happen, always fetch next instruction
 - Dynamic branch strategies
 - Keep a history of what has happened in the past
 - One bit stored for each history
 - 2 bits is the most common
 - Branch Prediction State Diagram
 - Branch History Table

- Three things to be kept in the table
 - Address of the branch instruction
 - Bits indicating branch history
 - Branch target information (where do we go if we branch?)
- Delayed Branching
 - In the RISC architecture, the instruction after a branch is executed even if the branch is taken